DEEP LEARNING BASED MARINE ECOSYSTEM MONITORING BY INTEGRATING YOLOv8l AND TINY U- NET ALGORITHM

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Abstract— Marine pollution, particularly oil spills and floating debris, poses a significant threat to aquatic ecosystems and requires efficient monitoring solutions. Traditional methods are often slow, expensive, and lack real-time responsiveness. This paper presents a deep learning-based marine ecosystem monitoring system that integrates YOLOv8l **for object detection and** Tiny U-Net **for pixel-level segmentation. The proposed framework enables accurate detection of floating waste and precise identification of oil spill regions from marine images. The system is trained using publicly available datasets, with class optimization applied to improve detection performance. Experimental results demonstrate that the YOLOv8l model achieves a high detection accuracy of** 95.8%**, while Tiny U-Net achieves approximately** 84% segmentation accuracy**. The combined approach provides a balanced solution in terms of accuracy and computational efficiency. The proposed system offers an automated, scalable, and reliable method for marine pollution monitoring, making it suitable for future real-time environmental applications.**

Keywords— Marine Pollution; YOLOv8l; Tiny U-Net; Deep Learning; Object Detection; Image Segmentation; Environmental Monitoring

I. INTRODUCTION

Marine ecosystems play a crucial role in maintaining environmental balance, supporting biodiversity, and sustaining human livelihoods. However, increasing levels of water pollution—particularly oil spills and floating debris—pose a severe threat to aquatic life and water quality. These pollutants not only disrupt marine habitats but also lead to long-term ecological and economic damage. Early detection and continuous monitoring are therefore essential to mitigate their impact. Conventional marine pollution monitoring methods rely on manual inspection, satellite imaging, or laboratory-based analysis. While these approaches provide valuable insights, they are often **time-consuming, expensive, and lack real-time responsiveness**. In addition, many existing systems depend on cloud-based processing and computationally heavy models, making them unsuitable for rapid and scalable deployment in practical scenarios. Recent advancements in deep learning, especially Convolutional Neural Networks (CNNs), have significantly improved the capability of automated image analysis systems. Object detection models such as YOLO (You Only Look Once) and segmentation models like U-Net have been widely adopted for environmental monitoring tasks. However, most existing research focuses on either detection or segmentation independently, without integrating both approaches into a unified system.

Furthermore, many implementations utilize complex architectures that require high computational resources, limiting their applicability in resource-constrained environments.To address these limitations, this work proposes a **novel hybrid deep learning framework** that integrates **YOLOv8l for object detection** and **Tiny U-Net for semantic segmentation** to achieve comprehensive marine pollution monitoring. The proposed system simultaneously detects floating debris using bounding boxes and identifies oil spill regions at the pixel level, enabling both **macro-level detection and micro-level analysis** within a single pipeline. A key contribution of this work lies in the use of **lightweight model architectures combined with a dual-task learning strategy**, which ensures high performance while maintaining computational efficiency. Unlike traditional approaches that treat detection and segmentation as separate tasks, the proposed system **synergistically combines both techniques**, enhancing the accuracy and reliability of pollution assessment. Additionally, the conversion of multi-class detection into a single-class problem improves model stability and detection consistency in complex marine environments.The implemented system demonstrates strong performance, achieving a detection accuracy of **95.8% using YOLOv8l** and approximately **84.9% segmentation accuracy using Tiny U-Net**. These results highlight the effectiveness of the proposed approach in balancing accuracy with efficiency. The framework is designed to be scalable and adaptable, making it suitable for future integration with real-time monitoring systems and environmental management applications.Marine pollution monitoring is particularly challenging due to **dynamic environmental conditions** such as varying lighting, water reflections, and wave motion, which make accurate detection difficult for traditional vision-based systems.The increasing availability of **unmanned aerial vehicles (UAVs) and marine cameras** has generated large volumes of visual data, creating a need for **automated and intelligent analysis systems**.Many existing deep learning models prioritize accuracy but fail to address **computational efficiency**, making them unsuitable for deployment in real-world environmental monitoring scenarios.The proposed system emphasizes a **balance between accuracy and efficiency**, which is critical for practical adoption in large-scale monitoring applications. The use of **data augmentation and optimized training strategies** improves model generalization, making the system robust to diverse environmental conditions.The proposed framework is designed with **modularity**, allowing easy extension to additional environmental parameters or integration with other intelligent systems.The system contributes toward the development of **smart environmental monitoring solutions**, aligning with global efforts in sustainability and ecological protection.

II. LITERATURE REVIEW

Marine pollution detection has become a critical research area due to the increasing environmental impact of oil spills and floating debris. Ensuring accurate, scalable, and automated monitoring systems remains a challenge, especially under dynamic marine conditions. Traditional image processing techniques relied on handcrafted features and threshold-based methods, which often fail to generalize across varying lighting conditions, water textures, and noise. As a result, deep learning-based approaches have gained significant attention for their ability to learn complex patterns directly from data. Early deep learning models for marine pollution detection primarily focused on **image classification techniques**, where Convolutional Neural Networks (CNNs) were used to determine whether an image contained pollution or not. For instance, K. Das et al. (2023) applied CNN-based classification for oil spill detection in satellite imagery. While this approach achieved reasonable accuracy, it lacked **spatial localization capability**, meaning it could not identify the exact location or extent of pollution within an image. This limitation reduces its effectiveness in real-world environmental monitoring, where precise information about pollution spread is essential. To overcome this limitation, researchers introduced **image segmentation techniques**, particularly encoder–decoder architectures such as U-Net. Gladkikh (2024) proposed a U-Net-based approach for detecting oil pollution using UAV and multispectral imagery. The model demonstrated strong performance in identifying polluted regions at the pixel level, enabling accurate estimation of spill coverage. However, segmentation-based approaches focus solely on region identification and do not provide information about **discrete objects such as floating debris**, which are equally important for comprehensive monitoring. In parallel, object detection models such as YOLO (You Only Look Once) have been widely used for real-time environmental monitoring. Indumathi et al. (2025) explored the use of YOLOv8 for detecting marine debris, demonstrating high accuracy and fast inference speed. YOLO-based models are particularly effective in identifying and localizing objects using bounding boxes, making them suitable for detecting floating waste. However, these approaches are limited to object-level detection and do not provide detailed insights into **pollution spread or affected regions**, which require pixel-level segmentation.Recent advancements have also explored the use of **transformer-based architectures** for marine pollution detection. Kang et al. (2024) compared CNN and transformer models using PlanetScope imagery and observed improved feature representation with transformers. However, these models require **large datasets and high computational resources**, making them less practical for efficient and scalable deployment, especially in resource-constrained environments.

Furthermore, Kalogirou et al. (2025) utilized Synthetic Aperture Radar (SAR) imagery combined with CNN models for oil spill detection. SAR-based methods offer robustness against weather conditions and lighting variations, but they involve **complex preprocessing steps and high computational overhead**, limiting their real-time applicability.A comprehensive review by Vasconcelos et al. (2023) highlighted key challenges in deep learning-based oil spill detection, including **high model complexity, lack of real-time processing, and limited integration of multiple analysis techniques**. The study emphasized the need for efficient models that can balance accuracy with computational performance while supporting practical deployment.

In addition to detection and segmentation techniques, recent research has also focused on improving the **robustness of marine pollution detection systems under real-world conditions**. Marine environments are highly dynamic, with variations in illumination, wave patterns, reflections, and occlusions. Traditional CNN-based models often struggle to generalize across such variations, leading to reduced performance in unseen scenarios. To address this, several studies have incorporated **data augmentation and preprocessing techniques**, such as contrast enhancement, denoising, and normalization, to improve model robustness. While these methods enhance performance, they increase preprocessing complexity and do not fully eliminate environmental challenges. Another important direction in the literature is the use of **multi-modal data sources**, including optical images, SAR imagery, and multispectral data. These approaches aim to improve detection accuracy by combining complementary information from different sensors. For instance, SAR-based methods are effective in detecting oil spills under low-visibility conditions, while optical images provide better spatial detail. However, integrating multi-modal data introduces challenges such as **data alignment, increased computational cost, and complex model design**, making such systems difficult to deploy in practical scenarios.

Recent works have also explored **real-time monitoring systems** using deep learning models. YOLO-based architectures have gained popularity due to their ability to perform detection in a single forward pass, enabling faster inference compared to traditional region-based methods. However, achieving real-time performance often requires **trade-offs between speed and accuracy**, particularly when using lightweight variants of detection models. While smaller models improve speed, they may compromise detection precision, especially for small or partially visible objects. In the context of segmentation, several variations of U-Net have been proposed to improve accuracy, including attention U-Net, residual U-Net, and multi-scale feature extraction techniques. These models enhance feature representation and boundary detection but significantly increase the number of parameters and computational requirements. As a result, there is a growing need for **compact segmentation models**, such as Tiny U-Net, that can deliver acceptable performance with reduced complexity.Another key challenge identified in the literature is the **lack of unified evaluation frameworks**. Different studies use varying datasets, metrics, and evaluation protocols, making it difficult to compare model performance consistently. While accuracy, precision, recall, and IoU are commonly used metrics, there is limited emphasis on **computational efficiency, inference time, and resource utilization**, which are critical for real-world deployment. Furthermore, many existing systems are designed as **standalone solutions**, focusing either on detection or segmentation without integrating additional analysis capabilities. For example, detection models can identify the presence of debris but cannot estimate pollution spread, while segmentation models can measure affected regions but cannot distinguish between different object types. This separation limits the overall effectiveness of environmental monitoring systems.

III.PROPOSED METHODOLOGY AND RESULTS

System Architecture: The proposed marine ecosystem monitoring system is designed as an integrated deep learning pipeline that combines **object detection and image segmentation** for comprehensive pollution analysis. The system processes marine images or video frames and produces both **bounding box detections for floating debris** and **pixel-level segmentation masks for oil spill regions**. The overall architecture consists of multiple interconnected modules, each responsible for a specific stage of data processing and analysis. The system operates through four primary modules: **Image Input Module**, **Preprocessing Module**, **Segmentation Module (Tiny U-Net)**, and **Detection Module (YOLOv8l)**, followed by an **Output and Visualization Module**. This modular design ensures flexibility, scalability, and efficient processing.

1.Image Input Module

The system accepts input in the form of **marine surface images or video streams** captured from cameras, drones, or satellite sources. These inputs may contain oil spills, floating debris, or other pollution elements present on the water surface.For video inputs, the system performs **frame extraction**, converting continuous video streams into individual frames for further processing. Each frame is treated as an independent input sample for detection and segmentation tasks.

2. Preprocessing Module

The preprocessing module standardizes and enhances the input data to improve overall model performance. Initially, all input images are resized to a fixed resolution of 640×640 to maintain uniformity across the dataset. Pixel values are normalized to ensure stable and efficient training, while noise reduction and image enhancement techniques are applied to improve feature clarity. Furthermore, to increase model robustness and generalization, several data augmentation techniques are incorporated during training. These include rotation (±5°), translation (0.1), scaling (0.3), horizontal flipping, and mosaic augmentation. Collectively, these preprocessing steps enable the model to effectively handle variations in lighting conditions, water reflections, and dynamic environmental factors present in marine ecosystems.

3. Oil Spill Segmentation Module (Tiny U-Net)

The preprocessed image is fed into the Tiny U-Net model to perform pixel-level segmentation of oil spill regions. Tiny U-Net is a lightweight encoder–decoder architecture specifically designed for efficient and accurate segmentation tasks. The encoder extracts hierarchical features from the input image, capturing important patterns such as texture and color variations, while the decoder reconstructs spatial information to generate precise segmentation masks. Additionally, skip connections are incorporated to retain fine-grained spatial details and improve localization accuracy. The final output of this module is a binary segmentation mask that highlights polluted regions, enabling effective estimation of oil spill coverage and the affected area.

4. Floating Debris Detection Module (YOLOv8l)

In parallel with the segmentation process, the same input image is processed using the YOLOv8l model for object detection. The model identifies floating debris by predicting bounding boxes around objects of interest, with each detection associated with a confidence score. To eliminate redundant detections, Non-Max Suppression (NMS) is applied, ensuring accurate and non-overlapping bounding boxes. To enhance detection stability and overall performance, the dataset was converted from a multi-class problem (five classes) into a single-class detection task, focusing solely on the presence of debris rather than class differentiation. The YOLOv8l model was trained using optimized hyperparameters, including 150 epochs, a batch size of 8, an image size of 640, the AdamW optimizer, and a learning rate of 0.0005. This configuration enabled the model to achieve a high detection accuracy of 95.8%.

5. Output and Visualization Module

The outputs generated from both the segmentation and detection models are combined to produce a comprehensive visualization of marine pollution. The segmentation masks are overlaid onto the original image to clearly highlight oil spill regions, while bounding boxes are drawn to indicate the presence of detected floating debris. The final output provides key analytical insights, including estimation of the polluted area, count of detected debris, and an intuitive visual representation for analysis. These processed results are both displayed and stored, enabling further evaluation, reporting, and potential integration into real-time monitoring systems.

Hybrid Detection–Segmentation Algorithm

The core of the proposed system is a hybrid analysis pipeline that integrates object detection and segmentation into a unified workflow for comprehensive marine pollution monitoring.

Input: Image or video frame I
Output: Segmentation mask M, bounding boxes B, final visualization V

1. **Begin**
2. Acquire input image/frame I
3. Resize I to fixed dimensions
4. Normalize pixel values
5. Apply image enhancement (if required)
6. $M \leftarrow$ Tiny U-Net(I)
7. Generate segmentation mask for oil spill regions
8. $B \leftarrow$ YOLOv8l(I)
9. Apply Non-Max Suppression (NMS) on B
10. Extract polluted regions from M
11. Count number of detected objects from B
12. Overlay M on original image
13. Draw bounding boxes B on image
14. Generate final visualization V
15. Save output
16. Display results
17. Return M, B, V
18. **End**

Experimental Setup and Evaluation:

The proposed system was evaluated using publicly available datasets for both object detection and segmentation tasks. A Roboflow marine debris dataset containing approximately 3700 images was used for detection, while the lados.v2i semantic mask dataset was used for oil spill segmentation. The detection dataset was simplified from five classes to a single class to improve model performance. All images were resized to 640×640, normalized, and augmented to enhance

generalization. The YOLOv8l and Tiny U-Net models were trained using optimized configurations, and their outputs were combined to perform comprehensive evaluation. The detailed configuration and performance of both models are presented in Table I and Table II.

Table I : YOLOv8l Detection Model Configuration and Performance

Parameters	**Value**
Dataset	Roboflow Marine Debris Dataset
Total Images	3700
Classes	1 (Converted from 5 classes)
Training Images	3200
Validation Images	515
Image Size	640 × 640
Epochs	150
Batch Size	8
Optimizer	AdamW
Learning Rate	0.0005
Task	Object Detection
Output	Bounding Boxes + Confidence Score
Accuracy	95.8%

Table II: Tiny U-Net Segmentation Model Configuration and Performance

Parameters	**Value**
Dataset	lados.v2i (Semantic Mask Dataset)
Input Type	Annotated Images with Masks
Image Size	640 × 640
Architecture	Encoder–Decoder (Tiny U-Net)
Task	Image Segmentation
Output	Binary Segmentation Mask
Training Type	Pixel-level Supervision
Augmentation	Rotation, Scaling, Translation, Flipping
Objective	Oil Spill Region Detection
Accuracy	84.9%

IV. RESULTS AND ANALYSIS

The proposed system was evaluated using both object detection and segmentation outputs. The results are analyzed using visual outputs, training curves, and quantitative performance metrics to assess model effectiveness under real-world marine conditions.

A.Detection Results (YOLOv8l)

The training and evaluation performance of the YOLOv8l model are analyzed using multiple graphical representations. The training loss curve (Fig. 1) shows a steady decrease in loss over epochs, indicating stable and efficient model convergence. The best epoch performance graph (Fig. 2) highlights the optimal training point based on evaluation metrics. In addition, the performance metrics curve (Fig. 3) illustrates trends in precision, recall, and mAP, confirming consistent detection capability. The evaluation results (Fig. 4) further validate the model's high accuracy and reliability.

The object detection results demonstrate that YOLOv8l effectively identifies floating debris by generating accurate bounding boxes along with confidence scores. The model achieved a high detection accuracy of **95.8%**, showing strong robustness under challenging conditions such as water reflections, varying illumination, and background noise. The conversion from multi-class to single-class detection significantly improved model stability and reduced misclassification errors.

Overall, these results confirm that YOLOv8l is highly effective for accurate and real-time detection of floating debris in marine environments.

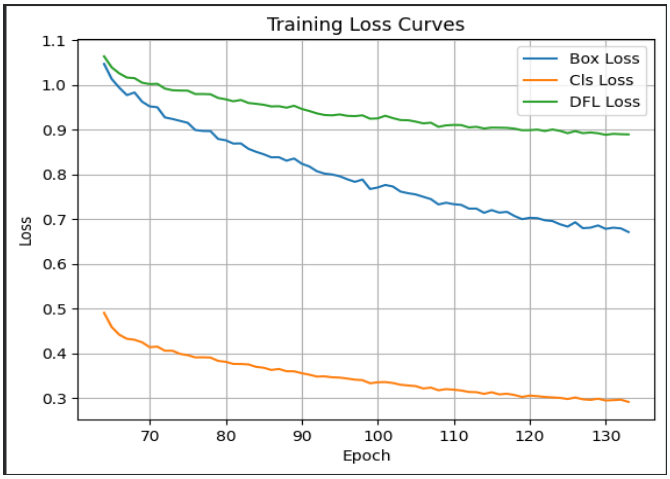

Fig. 1. YOLOv8l Training Loss Curve showing model convergence over epochs

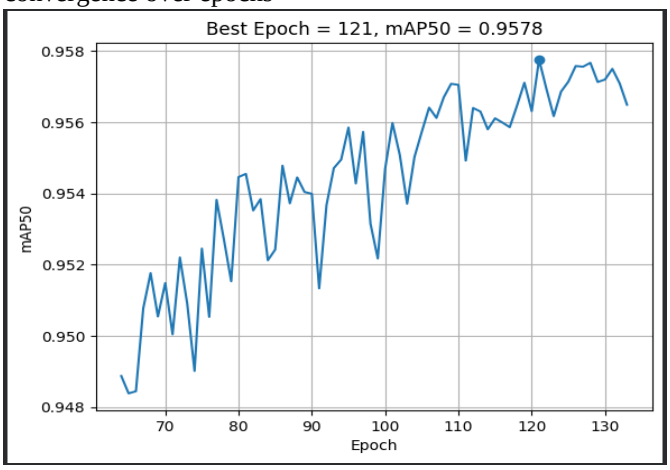

Fig.2: YOLOv8l Best Epoch Performance based on evaluation metrics

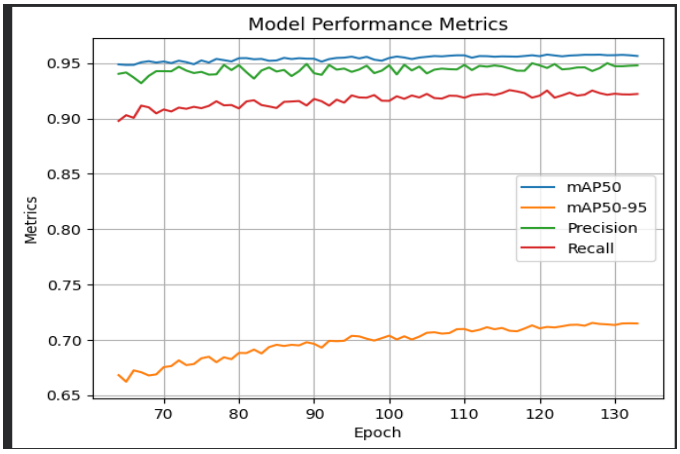

Fig.3: YOLOv8l Performance Metrics Curve (Precision, Recall, and mAP)

```
✅ Best Epoch Metrics:

Epoch        : 121
Precision (P): 0.9456
Recall (R)   : 0.9253
mAP50        : 0.9578
mAP50-95     : 0.7116

📉 Loss Values:
Box Loss     : 0.7022
Cls Loss     : 0.3045
Dfl Loss     : 0.8999
```

Fig.4: YOLOv8l Evaluation Results showing detection accuracy and metrics

B.Segmentation Results (Tiny U-Net)

The training and evaluation performance of the Tiny U-Net model are illustrated in Fig. 5–6. The evaluation metrics (Fig. 5) demonstrate the overall segmentation performance of the model, confirming its capability to accurately identify polluted regions.The segmentation outputs generated by Tiny U-Net are shown in Fig. 6, where the model produces pixel-level masks that effectively highlight oil spill regions in marine environments. It achieved an approximate segmentation accuracy of **84.9%**, indicating reliable performance in detecting polluted areas.

However, minor inaccuracies are observed near object boundaries due to the complexity of pixel-level classification and the visual similarity between oil and water textures. Despite these limitations, the model successfully captures the major spill regions, making it suitable for practical environmental monitoring applications.

```
# Get the last row of the DataFrame, which contains the final metrics after 50 epochs
final_metrics = results_df.iloc[-1]

# Extract and print the key accuracy metrics
print(f"Final mAP50-95: {final_metrics['metrics/mAP50-95(B)']:.4f}")
print(f"Final mAP50: {final_metrics['metrics/mAP50(B)']:.4f}")
print(f"Final Precision: {final_metrics['metrics/precision(B)']:.4f}")
print(f"Final Recall: {final_metrics['metrics/recall(B)']:.4f}")

Final mAP50-95: 0.5713
Final mAP50: 0.8498
Final Precision: 0.8553
Final Recall: 0.7541
```

Fig. 5. Tiny U-Net Evaluation Results showing segmentation performance metrics

Fig. 6. Tiny U-Net Segmentation Output showing oil spill regions

C.Performance Comparison

The overall performance of the proposed models is summarized in Table III.

Table III. Performance Of Proposed Models

Model	Task	Accuracy
YOLOv8l	Object Detection	95.8%
TinyU-Net	Segmentation	84.9%

The results clearly indicate that the YOLOv8l model achieves higher accuracy in object detection, primarily due to the simplified single-class representation and optimized training configuration. In contrast, the Tiny U-Net model provides effective segmentation performance with relatively lower computational complexity, making it suitable for pixel-level analysis of oil spill regions.

Overall, the combination of both models enables a balanced approach, delivering high detection accuracy along with reliable segmentation, thereby supporting comprehensive and efficient marine pollution monitoring.

D. Observations and Analysis

From the experimental results, it is observed that the YOLOv8l detection model demonstrates consistently high accuracy across diverse environmental conditions, including variations in lighting, reflections, and background complexity. The Tiny U-Net segmentation model effectively captures major oil spill regions, although minor boundary inconsistencies are observed due to the challenges of pixel-level classification and the visual similarity between oil and water textures. Furthermore, the use of data augmentation techniques significantly enhances model robustness and improves generalization across different marine scenarios. Overall, the proposed system achieves a balanced trade-off between accuracy and computational efficiency by integrating lightweight and optimized deep learning models.

V. CONCLUSION

This study demonstrates that the integration of object detection and image segmentation within a unified deep learning framework enables efficient and accurate marine pollution monitoring. The proposed system combines YOLOv8l for floating debris detection and Tiny U-Net for oil spill segmentation, providing both object-level and region-level analysis in a single pipeline.The hybrid approach achieves a strong balance between accuracy and computational efficiency, with YOLOv8l attaining **95.8% detection accuracy** and Tiny U-Net achieving approximately **84.9% segmentation accuracy**. The use of single-class detection and lightweight architectures enhances model stability while reducing computational complexity, making the system suitable for practical deployment.

Furthermore, the proposed framework supports automated, scalable, and reliable environmental monitoring by providing meaningful insights into pollution presence and spread. This can assist in effective decision-making for marine ecosystem management and protection.Future work will focus on enabling real-time deployment, improving segmentation performance using larger and more diverse datasets, and integrating additional environmental data sources to further enhance system robustness and real-world applicability.

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